

PIONEER JUNIOR COLLEGE
JC2 Preliminary Examinations
In preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

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INDEX
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BIOLOGY

9648/03

Application Paper and Planning Question

19 September 2016

Paper 3

2 hours

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your name, CT class and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graph or rough working.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.

All working for numerical answers must be shown.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question

For Examiner's Use	
1	
2	
3	
4 Planning	
5	
Total	

This document consists of **17** printed pages including the cover page and **1** blank page.

[Turn Over

Answer **all** questions.

- 1 Bloom's syndrome is a rare autosomal recessive disorder. It is characterized by short stature, predisposition to the development of cancer and genomic instability. Bloom's syndrome is caused by mutations in the BLM gene leading to mutated DNA helicase protein formation.

In the early 1980s, attempts were made to map the BLM gene on chromosome 15 using DNA markers. The relationship between the BLM gene and two DNA markers DM1 and DM2 were studied in a family with many affected children. DNA were extracted from these individuals and cut with two different restriction enzymes separately. The products of digestion were separated using gel electrophoresis and the RFLP banding patterns is shown in Fig. 1.1 below.

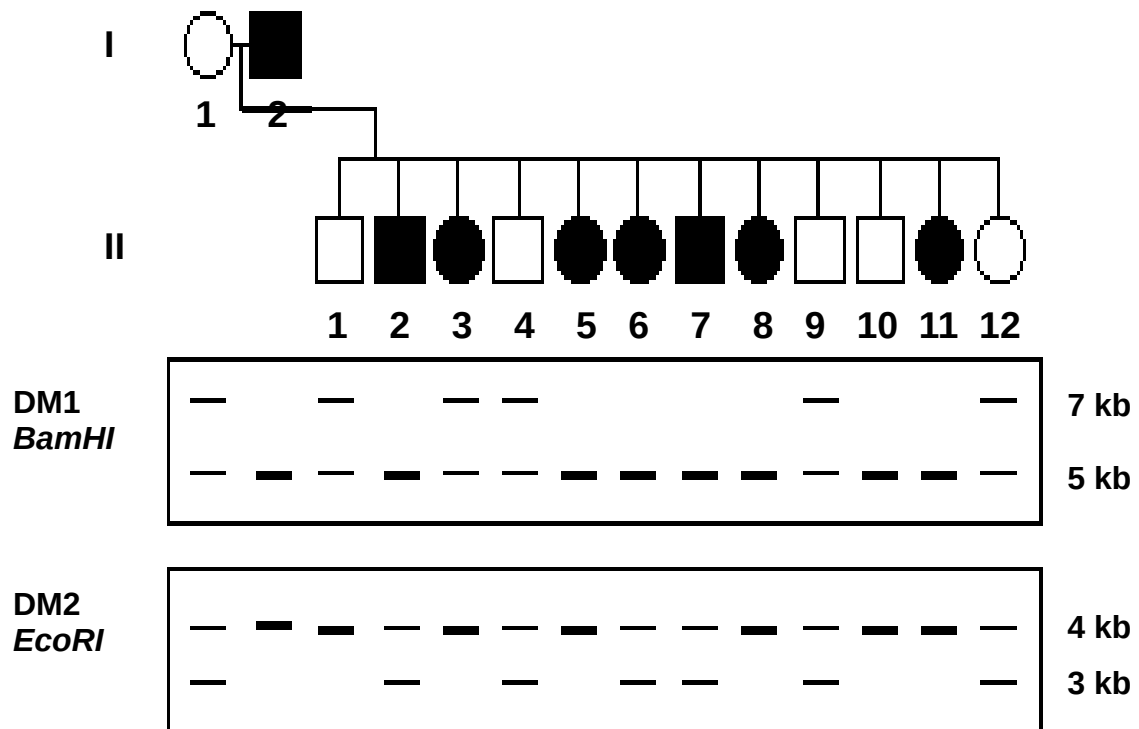


Fig. 1.1

(Note: *Bam*HI was used on DM1 and *Eco*RI on DM2)

Key

	Male with Bloom's syndrome		Female with Bloom's syndrome
	Normal male		Normal Female

- (a) Explain why genes are seldom used as markers for genetic testing.

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(b) What is meant by a restriction enzyme?

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(c) Explain the principles behind Restriction Fragment Length Polymorphism (RFLP) analysis.

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(d) Explain how gel electrophoresis can be used to distinguish between the different band patterns as observed in Fig. 1.1 using *Bam*HI.

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[Turn Over]

- (e) Using the information provided in Fig. 1.1, deduce and explain which marker is more suitable in detecting the presence of Bloom's syndrome.

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- (f) Using your understanding of linked genes, provide a possible explanation for your answer in (e).

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[Total: 14]

- 2 Severe Combined Immunodeficiency Disease (SCID) is the most severe of the immune deficiency diseases. SCID is a group of very rare, life-threatening diseases that are present at birth. Fig. 2.1 shows the pattern of inheritance of X-linked SCID in a family.

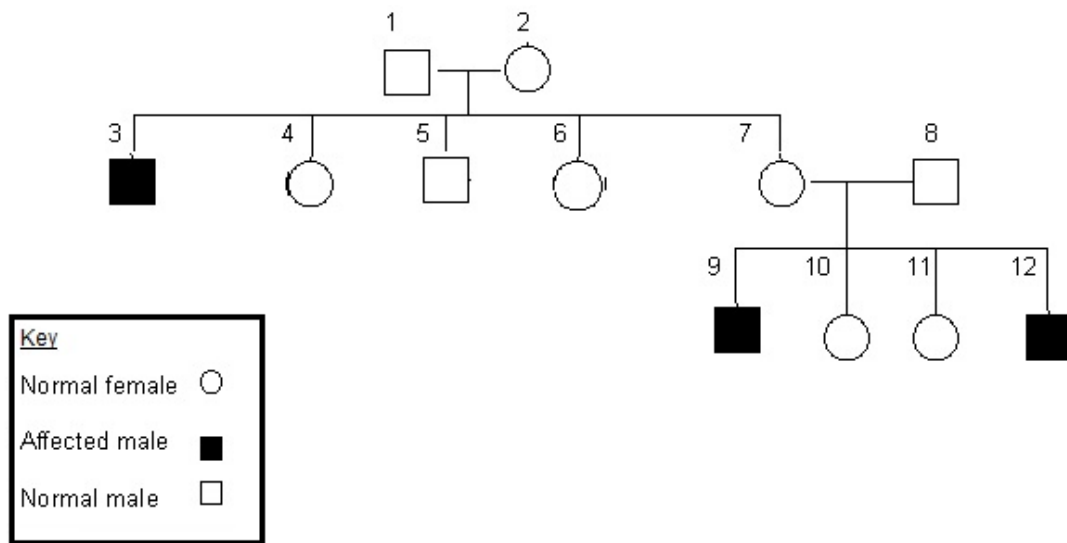


Fig. 2.1

- (a)** State and explain the nature of alleles that gives rise to SCID illustrated in Fig. 2.1.

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- (b)** Describe the genetic basis of X-linked SCID.

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(c) Give one reason why SCID is a suitable choice for gene therapy.

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Babies born with SCID have no defence against common infections and quickly become ill when the protection from maternal antibodies is lost. Gene therapy for SCID has been carried out using the general procedure shown in Fig. 2.2.

Step 1: Hematopoietic stem cells retrieved from baby's umbilical cord blood.



Step 2: Stem cells infected with harmless genetically engineered virus containing the normal, dominant allele.



Step 3: Stem cells take up normal allele.



Step 4: Stem cells transfused back into baby.



Step 5: Immune system develops T and B lymphocytes.

Fig. 2.2

(d) Using the information given, state **two** reasons why stem cells were used in this treatment.

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In 2002, two young patients with X-linked SCID were treated with their own blood stem cells that had been subjected to the same treatment as the earlier attempts at gene therapy.

The two different forms of SCID require different cells to be targeted for gene therapy. For X-linked SCID, only stem cells can be used while in autosomal recessive form of SCID (ADA), both T cells and stem cells could be used. If T cells are used, they will be induced to proliferate *in vitro* by treatment with relevant signalling molecules before infection with viral vectors.

- (e) From the information given and your knowledge of SCID, explain why different cells are used for different treatments.

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Fig. 2.3 shows the lymphocyte counts (both T & B) of two infant patients that underwent two separate types of gene therapy treatments for SCID. Patient A was treated with a retrovirus while Patient B was treated with an adenovirus. The lymphocyte count of the patients was monitored by a doctor at the end of every month for a period of 12 months. (The acceptable range of lymphocyte count for non-SCID patients is 5 to $10 \times 10^3/\text{mm}^3$).

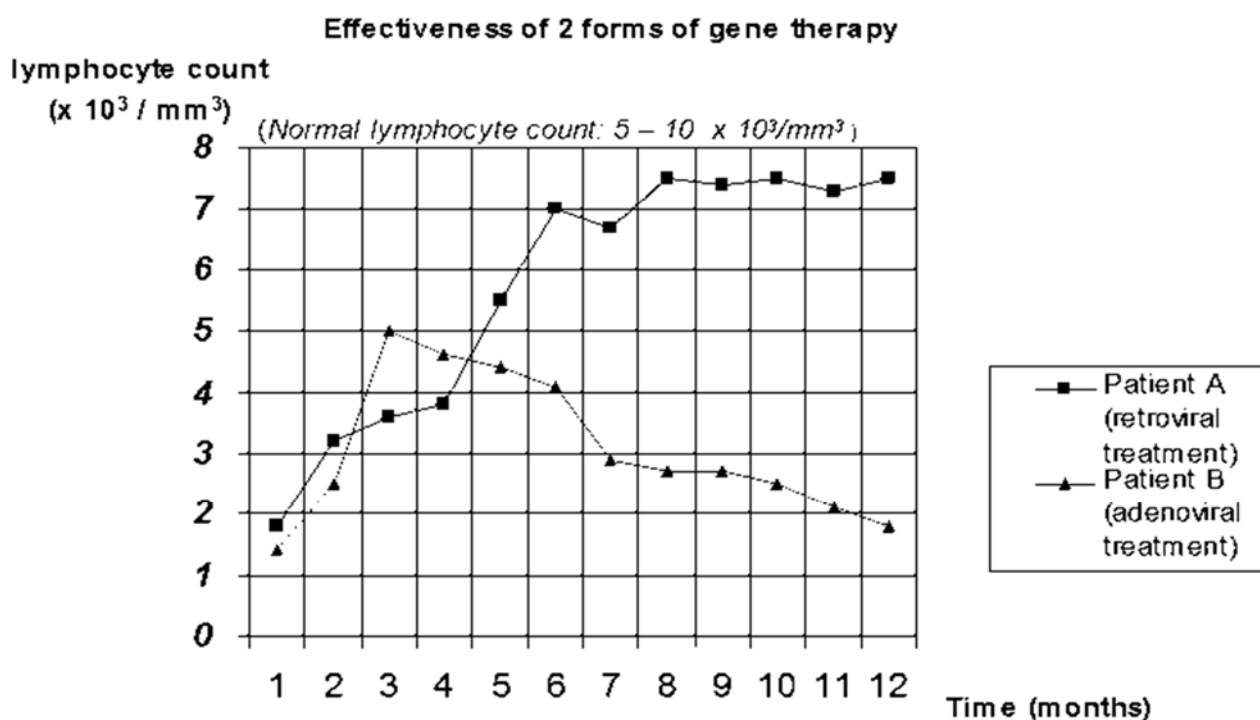


Fig. 2.3

[Turn Over

(f) With reference to Fig. 2.3,

(i) describe the trends observed in both patients' lymphocyte count.

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(ii) account for the difference in lymphocyte count at the end of 12 months.

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[Total: 14]

- 3 In 2012, permission was granted for a field trial in the UK of genetically modified wheat, *Triticum aestivum*. The wheat carries a gene, taken from peppermint plants, which results in the wheat leaves releasing a volatile, non-toxic chemical, (E)- β -farnesene (E β f), into the atmosphere.

E β f is not only produced by various species of plants. It is also secreted by aphids when they are disturbed by a predator.

Two experiments have been performed into the effect of E β f on the behaviour of aphids feeding on leaves in closed containers.

Experiment 1

Either 10 cm³ of air from a syringe that contained plant leaves that secrete E β f
or 10 cm³ of air from a syringe with normal leaves
was added to the containers of feeding aphids.

Experiment 2

Either 10 cm³ of air containing 50 ng of E β f
or 10 cm³ of air containing no E β f
was added to the containers of feeding aphids.

In both experiments, the number of aphids that stopped feeding and moved away from the food leaves was counted. The results are shown in Table 3.1.

Table 3.1

	Experiment 1		Experiment 2	
Air added to containers of feeding aphids	10 cm ³ air that had been in contact with leaves secreting E β f	10 cm ³ air that had not been in contact with leaves secreting E β f	10 cm ³ air containing 50 ng E β f	10 cm ³ air containing no E β f
Number of aphids in container	99	113	132	106
Number of aphids that stopped feeding and moved away from the food leaves	54	1	111	0

- (a) With reference to Table 3.1, describe the evidences which support the idea that E β f is an alarm signal for aphids.

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[Turn Over]

- (b) Other experiments show that E β f attracts predators of aphids, such as ladybirds. Suggest how growing genetically modified wheat secreting E β f could increase the yield of wheat.

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 [1]

- (c) Suggest why growing this genetically modified wheat might be acceptable to people who object to the growth of genetically modified insect-resistant maize or cotton.

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 [2]

In a separate experiment, Ti (Tumour-inducing) plasmid found in *Agrobacterium tumefaciens* was used to transfer DNA of interest into *T. triticum*. The transformed wheat protoplasts are stimulated to divide by mitosis to form calli and regenerated into whole plants by tissue culture techniques.

- (d) With reference to tissue culture techniques, describe how plant growth regulators can result in plantlet formation from calli.

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 [2]

Plasmids are small circles of DNA that can be found in many bacteria which can be used for cloning foreign genes in suitable bacteria.

The plasmid, pUC19, shown in Fig. 3.2 has been developed for specific purposes by genetic engineers.

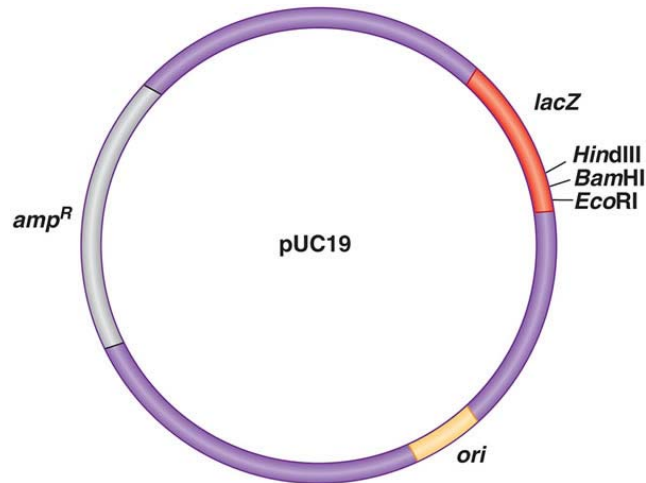


Fig. 3.2

pUC19 can be cleaved using restriction enzymes such as HindIII. HindIII can be found naturally in *Haemophilus influenzae*.

(e) What is the natural role of restriction enzymes in *H. influenzae*?

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(f) With reference to Fig. 3.2, explain how the cells which contain the recombinant plasmids are identified.

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[Total: 12]

[Turn Over]

4 Planning question

Dormant seeds have a very low rate of respiration. When water is absorbed by dormant seeds, growth hormones are activated. These hormones activate genes that code for enzymes that hydrolyse stored food reserves used in respiration allowing the seed to grow and germinate. The respiration rate can be measured by oxygen usage per unit mass using a respirometer.

You are required to investigate the respiration rate of germinating seeds of different mass.

You are provided with the following materials and apparatus which you must use:

- Mung bean seeds of different mass that have been soaked in water for 24 hours
- Blue dye
- Soda lime pellets 
- Glass beads
- Rubber tubing connected to capillary tube of 4mm diameter
- Syringes
- Electronic weighing balance
- Scale paper

You may select from the following apparatus and use appropriate additional apparatus:

- Normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc
- Forceps
- Ruler
- Stopwatch
- Thermometer

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s), if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

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H2 Bio/9648/Prelim/P3

5 Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

(a) Describe PCR and explain the advantages and limitations of this technique. [8]

(b) Describe how plant tissue culture is used to clone plant cells. Explain the scientific reasons for each step in the process. [8]

(c) Distinguish between a genomic DNA library and a cDNA library. [4]

[Total: 20]

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